

Boris Houska

ShanghaiTech University
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[Personal Website](#)

Experience

Associate Professor, ShanghaiTech University, 2020–present
Visiting Professor, IMT School for Advanced Studies Lucca, Italy, Summer 2024
Assistant Professor, ShanghaiTech University, 2014–2020
Academic Visitor, CPSE, Imperial College London, Summer 2018
Visiting Scholar, EECS, University of California, Berkeley, Winter 2016
Visiting Professor, Institute for Microsystems Engineering, University of Freiburg, Summer 2014
Visiting Professor, Freiburg Institute for Advanced Studies, University of Freiburg, Summer 2014
Research Faculty, Department of Automation, Shanghai Jiao Tong University, 2013–2014
Postdoctoral Researcher, CPSE, Imperial College London, 2012–2013
Postdoctoral Researcher, Optimization in Engineering Center, KU Leuven, 2011

Education

Ph.D. in Electrical Engineering, KU Leuven, 2011
Diploma in Mathematics, University of Heidelberg, 2007

Honors and Awards

- Scientific Advisory Board Member, EU Project ELO-X, 2021–2024.
- SIST Excellence in Research Award, ShanghaiTech University, 2019.
- ShanghaiTech Excellent Professor Award, ShanghaiTech University, 2016.
- SIST Excellence in Research Award, ShanghaiTech University, 2016.
- FRIAS Fellowship (for visiting professors), Freiburg Institute for Advanced Studies, University of Freiburg, 2014.
- ICCOPT Best Paper Award for a Young Researcher in Continuous Optimization (Finalist, Top 3) for the paper “Nonlinear Robust Optimization via Sequential Convex Bilevel Programming” by B. Houska and M. Diehl, Mathematical Programming, Series A, 2013.
- Marie-Curie Fellowship for the project “Next Generation Algorithms for Robust and Global Optimization of Dynamic Systems”, 2012.

Research Projects and Leadership Roles

- External Partner, EU project ELO-X on “Embedded Learning and Optimization for the Next Generation of Smart Industrial Control Systems”; 2021–2024.
- PI, Bilateral Grant Agreement (Sino-German Mobility Program; together with Prof. Timm Faulwasser, Prof. Karl Worthmann, Prof. Matthias Müller) for the project “Distributed Optimization and Control of Networked Energy Systems”, 2020–2022.
- PI, Bilateral Grant Agreement (Slovakia-China, together with Prof. Radoslav Paulen) for the project “Verified Estimation and Control of Chemical Processes”, 2018–2020.
- Full member of the Scientific Network on “External Regulation of Mobile Robots using Model Predictive Control: Beyond Set Point Stabilization” funded by German Research Foundation (DFG), 2017–2018.
- PI, Bilateral Grant Agreement (Slovakia-China, together with Prof. Michal Kvasnica) for the project “Robust Model Predictive Control meets Robotics”, 2015–2017.
- PI, NSFC Grant for the project “Embedded Optimization Algorithms for Economic Model Predictive Control”, National Natural Science Foundation China, 2015–2018.
- Associated member of the Scientific Network on “External Regulation of Mobile Robots using Model Predictive Control: Set Point Stabilization, Path Following, and Distributed Control” funded by German Research Foundation (DFG), 2015–2016.
- PI, Marie-Curie project (FP7) “Next Generation Algorithms for Robust and Global Optimization of Dynamic Systems”, 2012.
- Various institutional grants (Imperial College London (2013), Shanghai Jiao Tong University (2013/14), ShanghaiTech University (2014–present)).

Selected Invited Lectures

1. Invited presentation at the Minisymposium on Operator Methods in Engineering, WCCM-ECCOMAS, Munich, Germany, 2026.
2. Institute for Mathematical and Statistical Innovation, University of Chicago, USA, 2025.
Video available at [IMSI WEBSITE](#)
3. Slovak University of Technology in Bratislava, Slovakia, 2025.
4. School of Computation, Information and Technology, TU München, Germany, 2025.
5. Institute of Control Systems, TU Hamburg, Germany, 2025.
6. CDC Workshop on “Data-driven modelling, analysis, and control using the Koopman operator”, Milano, Italy, 2024.
7. IMT School for Advanced Studies Lucca, Italy, 2023.
8. Automatic Control Laboratory, EPFL, Switzerland, 2023.
9. School of Computation, Information and Technology, TU München, Germany, 2023.
10. Department of Mathematics, TU Ilmenau, Germany, 2023.
11. Department of Mathematics, University of Bayreuth, Germany, 2023.

12. **Semi-Plenary Lecture**, 25th MTNS Conference, Bayreuth, Germany, 2022.
This is the prime conference on all mathematical aspects in systems and control theory.
13. **Invited Tutorial Lecture**, American Control Conference, Atlanta, 2022.
14. Invited Presentation, IFAC Virtual Seminar (TC 2.4 on Optimal Control), 2022.
15. IMTEK, Faculty of Engineering, University of Freiburg, Germany, 2021.
16. Slovak University of Technology in Bratislava, Slovakia, 2019.
17. IMTEK, Faculty of Engineering, University of Freiburg, Germany, 2019.
18. Karlsruhe Institute of Technology, Germany, 2019.
19. **Keynote Lecture**, CPSE Summer School, Imperial College London, UK, 2018.
20. Slovak University of Technology in Bratislava, Slovakia, 2018.
21. International Symposium on Mathematical Programming, France, 2018.
22. 14th Viennese Conference on Optimal Control and Dynamic Games, Vienna, 2018.
23. Fourth Symposium on Future Innovations in Process Systems Engineering, Greece, 2018.
24. **Keynote Lecture** (in the form of a workshop), 21st International (IFAC) Conference on Process Control, Štrbské Pleso, Slovakia, 2017.
25. Institute for Automation Engineering, Otto von Guericke University Magdeburg, 2018.
26. INFORMS Annual Meeting, Houston, USA, 2017.
27. Automatic Control, Automation and Mechatronics Division, Chalmers, 2017.
28. Bremen Institute of Industrial Technology, University of Bremen, 2017.
29. Slovak University of Technology in Bratislava, Slovakia, 2017.
30. Hybrid Systems Group Seminar, EECS, University of California, Berkeley, 2017.
31. Joint Annual Meeting of the International Association of Applied Mathematics and Mechanics, Germany, 2017.
32. Institute for Microsystems Engineering, University of Freiburg, 2016.
33. Institute for Systems Theory and Automatic Control, University of Stuttgart, 2016.
34. 5th International Conference on Continuous Optimization, Tokyo, Japan, 2016.
35. Third Symposium on Future Innovations in Process Systems Engineering Greece, 2016.
36. Slovak University of Technology in Bratislava, Slovakia, 2016.
37. Workshop on Robust Optimization-Based Control and Planning for Legged Robots, IEEE International Conference on Robotics and Automation, Stockholm, 2016.
38. Workshop on Economic and Distributed Model Predictive Control, Swiss Federal Institute of Technology in Lausanne, Switzerland, 2016.
39. 1st Symposium on Robotics and Model Predictive Control: Path Following, BIBA, University of Bremen, Germany, 2015.
40. Department of Chemical Engineering, Carnegie Mellon University, USA, 2015.
41. International Symposium on Mathematical Programming, Pittsburgh, USA, 2015.
42. Magdeburg Lectures on Optimization and Control, University of Magdeburg, 2014.

43. FRIAS Workshop on Robust Optimization and Control, University of Freiburg, 2014.
44. SIAM Conference on Optimization, San Diego, USA, 2014.
45. OPTEC Mini Symposium, KU Leuven, Belgium, 2014.
46. AIChE Annual Meeting, Atlanta, USA, 2014.
47. ShanghaiTech University, Shanghai, China, 2014.
48. **Plenary Lecture**, Best Paper Award Session, International Conference on Continuous Optimization (ICCOPT), Lisbon, Portugal, 2013.
49. AIChE Annual Meeting, San Francisco, USA, 2013.
50. 4th International Conference on Continuous Optimization, Lisbon, Portugal, 2013.
51. **Plenary Lecture**, 2nd International Workshop on Mathematics and Algorithms for Computer-Aided Manufacturing, Engineering and Numerical Control (MAMENC), Beijing, China, 2012.
52. INFORMS Annual Meeting, Phoenix, USA, 2012.
53. Computational Management Science Conference, London, UK, 2012.
54. International Symposium on Mathematical Programming, Berlin, Germany, 2012.
55. Workshop on Modeling, Simulation, and Optimization of Uncertain Systems, Heidelberg, Germany, 2012.
56. QUADS Seminar, Imperial College London, UK, 2012.
57. Benelux Meeting on Systems and Control, Nijmegen, Netherlands, 2012.
58. International Conference on Multiple Criteria Decision Making, Jyväskylä, Finland, 2011.
59. SIAM Conference on Optimization, Darmstadt, Germany, 2011.
60. International Airborne Windenergy Conference, Leuven, Belgium, 2011.
61. School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2009.
62. SIAM Conference on Optimization, Boston, USA, 2009.
63. Belgian-French-German Conference on Optimization, Leuven, Belgium, 2009.
64. Conference on Modeling, Simulation and Optimization of Complex Processes, Heidelberg, Germany, 2008.
65. OPTEC Workshop on Distance Measures and Eigenvalue Optimization, Leuven, Belgium, 2008.
66. Workshop on Robust MPC, Leuven, Belgium, 2008.

Poster Awards

- CAST Directors' Award for the poster "Higher-Order Inclusions of Nonlinear Systems" by B. Chachuat, J. Rajyaguru, M.E. Villanueva, and B. Houska, 2015.
- Best Poster Award (3rd prize) for the poster "Efficient numerical methods for computation of open-loop stable kite orbits." by J. Sternberg, B. Houska, M. Diehl, Airborne Wind Energy Conference (AWEC), 2011.

URL Links

- Personal Website: <https://boris-houska.github.io>
- Orcid ID: <https://orcid.org/0000-0002-6761-239X>
- Google Scholar: scholar.google.com/citations?user=FetRI5cAAAAJ&hl=en
- Scopus: <https://www.scopus.com/authid/detail.uri?authorId=23008711800>
- ResearchGate: https://www.researchgate.net/profile/Boris_Houska

Advising

Postdocs

- Main supervisor of **Mario E. Villanueva**, ShanghaiTech University, 2017–2022.
 - Winner of the 2016 Dudley Newitt Prize
 - Winner of the 2018 SIST Excellent Postdoc Award
 - Winner of the 2019 SIST Excellent Postdoc Award
 - Now Associate Professor at IMT Lucca

Students

- During my time at ShanghaiTech University I have been the main supervisor of more than 10 graduate students, including Master's Degrees and PhD tracks, and a long list of undergraduate students.

Professional Service

Technical Meetings—Leadership Roles

- Co-organizer of the invited session "Trends in Optimization for Power Systems", The 61st IEEE Conference on Decision and Control, 2022.
- Co-organizer of the invited session "Advances in set-based and stochastic methods for estimation and control of nonlinear process systems", 11th IFAC International Symposium on Advanced Control of Chemical Processes, 2021.
- Organizer of the IFAC Pre-Conference Workshop "Machine Learning meets Model Based Control" (together with Ali Mesbah, Berkeley), 21st IFAC World Congress, July, 2020.
With more than 300 participants this workshop was among the most well attended pre-conference events in the history of IFAC World Congress.
- Organizer of the "Workshop on Set-Based Methods in Estimation and Control", 12th IFAC Symposium on Dynamics and Control of Process Systems, April 23–26, 2019.
- Co-organizer of the invited session "Advances in Stochastic and Set-Based Control and Estimation", 12th IFAC Symposium on Dynamics and Control of Process Systems, April 23–26, 2019.
- Organizer of the "International Workshop on Advanced Methods for Control and Estimation of Dynamic Systems (AMCEDS 2018)", ShanghaiTech University, Shanghai, China, July 23, 2018.
- Organizer of the Scientific Network Meeting "Regulation of Mobile Robots using MPC: Beyond Set Point Stabilization", Chalmers University of Technology, Gothenburg, Sweden, September 4–6, 2017.
- Organizer of the "2015 ShanghaiTech Workshop on Model Predictive Control", School of Information Science and Technology, ShanghaiTech, China, December 11, 2015.
- Organizer of the "FRIAS Workshop on Robust Optimization and Control", Freiburg Institute for Advanced Studies, University of Freiburg, Germany, August 20, 2014.

- Co-organizer of the OPTEC Workshop on Distance Measures and Eigenvalue Optimization, June 19-20, Leuven, 2008.
- Co-organizer of the International Workshop on Modelling and Optimization of Power Generating Kites, January 30, Leuven, 2007.

Technical Program Committee

- International program committee member, 35th European Symposium on Computer Aided Process Engineering, ESCAPE, 2025.
- International program committee member, 14th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems, DYCOPS, 2025.
- International program committee member, 8th IFAC Conference on Nonlinear Model Predictive Control, IFAC-NMPC, 2024.
- International program committee member, 13th Dynamics and Control of Process Systems, DYCOPS-CAB, 2022.
- International program committee member, 58th IEEE Conference on Decision and Control, December 11–13, 2019.
- Member of the Technical Program Committee on Process Control, IEEE Control Systems Society, 2018–present.
<http://process-control.ieeecss.org/industrial-home>
- International program committee member, 12th IFAC Symposium on Dynamics and Control of Process Systems, April 23–26, 2019.
- International program committee member, 6th IFAC International Conference on Nonlinear Model Predictive Control, August 19–22, 2018.
- International program committee member, 15th IEEE International Symposium on Safety, Security, and Rescue Robotics, October 11–13, 2017.
- International program committee member, Airborne Wind Energy Conference, October 5–6, 2017.

Conference Session Chair/Co-chair

- Chair of the session *Robust Control I*, 63rd IEEE Conference on Decision and Control, Milano, Italy, December, 2024.
- Area chair (for the area “Model Based Control”), 12th IFAC Symposium on Advanced Control of Chemical Processes (ADCHEM), Toronto, July, 2024.
- Area chair (for the area “Optimization and Computing in Control”), 24th International (IFAC) Conference on Process Control, June, 2023.
- Area chair (for the area “Optimization and Computing in Control”), 23rd International (IFAC) Conference on Process Control, June, 2021.
- Co-chair of the session *Predictive Control for Nonlinear Systems*, 58th IEEE Conference on Decision and Control, Nice, France, December, 2019.

- Chair of the session *Predictive Control of Nonlinear Systems I*, 57th IEEE Conference on Decision and Control, Florida, USA, December, 2018.
- Area chair (for the area “optimization”), 6th IFAC International Conference on Nonlinear Model Predictive Control, August, 2018.
- Chair of the outbreak session *Open problems, tools, and methods to handle and tackle uncertainty in large scale systems*, Fourth Symposium on Future Innovations in Process Systems Engineering (FIPSE’18), Chakidiki, Greece, June 25–27, 2018.
- Organizer and chair of the invited session *Global Optimization*, INFORMS Annual Meeting, Houston, USA, October, 2017.
(together with Dr. Mario E. Villanueva)
- Chair of the session “Sparse Representation”, ShanghaiTech Symposium on Data Science, Shanghai, June 2015.
- Chair of the session “Large Scale Parallel and Distributed Optimization”, ShanghaiTech Symposium on Data Science, Shanghai, June 2015.
- Chair of the invited session *Topics in Global Optimization*, INFORMS Annual Meeting, Phoenix, USA, 2012.
- Chair of the session *Robust Nonlinear Control I*, 7th IFAC Symposium on Robust Control Design, Aalborg, Denmark, 2012.
- Co-chair of the session *Robustness Analysis*, 7th IFAC Symposium on Robust Control Design, Aalborg, Denmark, 2012.

Expert Reviewer for Funding Agencies

- Expert reviewer for Research Foundation Flanders (FWO), 2017.
- Expert reviewer for German Research Foundation (DFG), 2016/17 and 2020–present.

Editorial Activities

- Associate Editor, IFAC Conference on Nonlinear Model Predictive Control, 2024.
- Associate Editor, IFAC Symposium on Advanced Control of Chemical Processes, 2024.
- Associate Editor, IFAC Conference on Process Control, 2023.
- Associate Editor, IFAC Conference on Process Control, 2021.
- Associate Editor at Large, IEEE Conference on Decision and Control, 2019.
- Subject Editor, Optimal Control Applications & Methods, 2018–2022.
- Associate Editor, IFAC NMPC (Conference Proceedings), 2018.

Technical Reviewer

- Selected journals: Automatica, IEEE TAC, SIOPT, SICON, Math. Prog., JOGO, etc.
- Selected international control conferences: CDC, ECC, ACC, IFAC World Congress, etc.

Opponent Services

- PhD Jury Committee Member, Alex Andriën, TU Eindhoven, 2022.
- External assessor (member of Ph.D. jury committee) of Dries Telen, KU Leuven, 2014.
- Co-examiner of Benedikt Schleusener's Bachelor Defense, University of Freiburg, 2014.

Teaching

ShanghaiTech University (Institute of Mathematical Sciences)

- Linear Algebra.
Fall 2024, Fall 2025.
- Mathematical Modeling.
Spring 2026.
- Numerical Analysis.
Fall 2022, Spring 2024, Spring 2025.

ShanghaiTech University (School of Information Science and Technology)

- Mathematical Control Theory.
Spring 2023.
- Introduction to Control.
Spring 2017, Fall 2017, Fall 2018, Fall 2019, Fall 2020, Spring 2022.
- Numerical Analysis.
Fall 2014, Fall 2015, Spring 2017, Fall 2017, Fall 2018, Fall 2019, Fall 2020, Spring 2022.
- Optimal Control.
Summer 2016.
- Linear Systems.
Spring 2015 and Spring 2016.
- Linear Algebra.
As substitute teacher (2 x 2h): Fall 2017.
- Introduction to Information Science and Technology.
As assistant teacher: Spring 2015, Spring 2016, Spring 2017, Spring 2018, and Spring 2019.

University of Freiburg

- Organizer of the “TEMPO-Course on Robust Optimal Control” (together with S. Diamond, Stanford), September 3-4, University of Freiburg, 2014.

Shanghai Jiao Tong University

- Assistant teacher of the course “Basic Principles of Automation” (together with Wei Wang), Fall 2013.

University of Trento

- Co-organizer and teacher of the “Nonlinear Dynamic Optimization Course” (together with M. Diehl), July 4-8, Trento, 2011.

Norwegian University of Science & Technology

- Co-organizer and teacher of the “Numerical Optimal Control Course” (together with M. Diehl), June 7-11, Trondheim, 2010.

KU Leuven

- Organizer of the OPTEC Tutorial Course on Automatic Control and Dynamic Optimization (together with M. Diehl and H.J. Ferreau), July 14-15, Leuven, 2011.
- Co-organizer and teacher of the ATHENS Course on Embedded and Convex Optimization for Control (together with S. Boyd and M. Diehl), March 15-19, Leuven, 2010.
- Co-organizer and teacher of the ATHENS Course on Numerical Optimal Control Algorithms and Applications in Renewable Energy Systems (together with H.J. Ferreau and M. Diehl), November 16-20, Leuven, 2009.

Publications

Summary

Dr. Houska has over 58 journal papers published or in press, one of which has won a best paper award. He has published 62 peer-reviewed conference papers and 3 book chapters. His publications have been cited over 6000 times in Google Scholar with an h-index of 35 as of March 2026. Links to his publications can be found at scholar.google.com/citations?user=FetRI5cAAAAJ&hl=en

Journal Papers

1. N. Hoischen, P. Bevanda, S. Sosnowski, S. Hirche and B. Houska. Data-driven stochastic optimal control in reproducing kernel Hilbert spaces, *Engineering Applications of Artificial Intelligence*, 2026. (in print)
2. Y. Zha, M.E. Villanueva, B. Houska, B. Chachuat. Relaxation via separable estimators: arithmetic and implementation. *Journal of Global Optimization*, 2026. (provisionally accepted)
3. B. Houska, M.A. Müller, M.E. Villanueva. Control Lyapunov function design via configuration-constrained polyhedral computing. *Automatica*, Volume 188, 112896, 2026.
4. B. Houska. Intrinsic separation principles. *Automatica*, Volume 183, 112661, 2026.
5. B. Houska, M.A. Müller, M.E. Villanueva. Polyhedral control design: Theory and methods. *Annual Reviews in Control*, Volume 60, 100992, 2025.
6. B. Houska. Convex operator-theoretic methods in stochastic control. *Automatica*, Volume 177, 112274, 2025.
7. Y. Jiang, K. Fedorova, J. Su, J. Oravec, B. Houska, C.N. Jones. Fast and lightweight: a real-time parallelizable MPC for embedded systems. *European Journal of Control*, Volume 83, 101217, 2025.
8. B. Houska, M.A. Müller, M.E. Villanueva. On stabilizing terminal costs and regions for configuration-constrained Tube MPC. *IEEE Control Systems Letters*, Volume 8, pp. 1961–1966, 2024.
9. F. Badalamenti, S.K. Mulagaleti, A. Bemporad, B. Houska, M.E. Villanueva. Configuration-constrained Tube MPC for tracking. *IEEE Control Systems Letters*, Volume 8, pp. 1174–1179, 2024.
10. M.E. Villanueva, M.A. Müller, B. Houska. Configuration-constrained tube MPC. *Automatica*, Volume 163:111543, 2024.
11. X. Gao, M.E. Villanueva, B. Houska. Multi-symmetric Lyapunov equations. *IEEE Control Systems Letters*, Volume 7, pp. 490–495, 2023.
12. J. Shi, Y. Jiang, J. Oravec, B. Houska. Parallel MPC for linear systems with state and input constraints. *IEEE Control Systems Letters*, Volume 7, pp. 229–234, 2023.

13. F. Wu, M.E. Villanueva, B. Houska. Ambiguity tube MPC. *Automatica*, Volume 146:110648, 2022.
14. A. Engelmann, Y. Jiang, H. Benner, R. Ou, B. Houska, T. Faulwasser. ALADIN- α – An open-source MATLAB toolbox for distributed non-convex optimization. *Optimal Control Applications & Methods, Special Issue Article*, Volume 43, pp. 4–22, 2022.
15. A. Murray, T. Faulwasser, V. Hagenmeyer, M.E. Villanueva, B. Houska. Partially distributed outer approximation. *Journal of Global Optimization*, Volume 80, pp. 523–550, 2021.
16. M.E. Villanueva, C.N. Jones, B. Houska. Towards global optimal control via Koopman lifts. *Automatica*, Volume 132:109610, 2021.
Editors' Choice — Automatica:
<https://www.journals.elsevier.com/automatica/editors-choice/october-2021-automatica>
17. Y. Jiang, P. Sauerteig, B. Houska, K. Worthmann. Distributed optimization using ALADIN for MPC in smart grids, *IEEE Transactions on Control Systems Technology*, Volume 29(5), pp. 2142–2152, 2021.
18. Y. Jiang, J. Oravec, B. Houska, M. Kvasnica. Parallel MPC for Linear Systems with Input Constraints. *IEEE Transactions on Automatic Control*, Volume 66(7), pp. 3401–3408, 2021.
19. Y. Jiang, D. Kouzoupis, H. Yin, M. Diehl, B. Houska. Decentralized optimization over tree graphs. *Journal of Optimization Theory and Applications*, Volume 189, pp. 384–407, 2021.
20. A. Engelmann, Y. Jiang, B. Houska, T. Faulwasser. Decomposition of non-convex optimization via bi-level distributed ALADIN. *IEEE Transactions on Control of Network Systems*, Volume 7(4), pp. 1848–1858, 2020.
21. Y. Jiang, J. Su, Y. Shi, B. Houska. Distributed optimization for massive connectivity, *IEEE Wireless Communication Letters*, Volume 9(9), pp. 1412–1416, 2020.
22. M.E. Villanueva, E. De Lazzari, M.A. Müller, B. Houska. A set-theoretic generalization of dissipativity with applications in Tube MPC. *Automatica*, Volume 122:109179, 2020.
23. X. Feng, M.E. Villanueva, B. Houska. Backward-forward reachable set splitting for state-constrained differential games. *Automatica*, Volume 111:108602, 2020.
24. M.E. Villanueva, B. Houska. On stochastic linear systems with zonotopic support sets. *Automatica*, Volume 111:108652, 2020.
25. B. Houska, B. Chachuat. Global optimization in Hilbert space. *Mathematical Programming, Series A*, Volume 173, pp. 221–249, 2019.
26. A. Engelmann, Y. Jiang, T. Mühlpfordt, B. Houska, T. Faulwasser. Towards distributed OPF using ALADIN. *IEEE Transactions on Power Systems*, Volume 34(1), pp. 584–594, 2019. **150+ citations in GS**
27. X. Feng, B. Houska. Real-time algorithm for self-reflective model predictive control. *Journal of Process Control*, Volume 65, pp. 68–77, 2018.

28. B. Houska, J.C. Li, B. Chachuat. Towards rigorous robust optimal control via generalized high-order moment expansion. *Optimal Control, Applications and Methods*, Volume 39(2), pp. 489–502, 2018.
29. R. Quirynen, S. Gros, B. Houska, M. Diehl. Lifted collocation integrators for direct optimal control in ACADO toolkit. *Mathematical Programming Computation*, Volume 9(4), pp. 527–571, 2017.
30. B. Houska, D. Telen, F. Logist, J. Van Impe. Self-reflective model predictive control. *SIAM Journal on Control and Optimization*, Volume 55(5), pp. 2959–2980, 2017.
31. B. Houska, M.A. Müller. Cost-to-travel functions: a new perspective on optimal and model predictive control. *Systems & Control Letters*, Volume 106, pp. 79–86, 2017.
32. R. Quirynen, B. Houska, M. Diehl. Efficient symmetric Hessian propagation for direct optimal control. *Journal of Process Control*, Volume 50, pp. 19–28, 2017.
33. M.E. Villanueva, R. Quirynen, M. Diehl, B. Chachuat, B. Houska. Robust MPC via min–max differential inequalities. *Automatica*, Volume 77, pp. 311–321, 2017.
130+ citations in GS
34. D. Telen, B. Houska, M. Vallerio, F. Logist, J. Van Impe. A study of integrated experiment design for NMPC applied to the Droop model. *Chemical Engineering Science*, Volume 160, pp. 370–383, 2017.
35. J. Rajyaguru, M.E. Villanueva, B. Houska, B. Chachuat. Chebyshev models arithmetic for factorable functions. *Journal of Global Optimization*, Volume 68(2), pp. 413–438, 2017.
36. B. Houska, A. Mohammadi, M. Diehl. A short note on constrained linear control systems with multiplicative ellipsoidal uncertainty. *IEEE Transactions on Automatic Control*, Volume 61(12), pp. 4106–4111, 2016.
37. B. Houska, J. Frasch, M. Diehl. An augmented Lagrangian based algorithm for distributed non-convex optimization. *SIAM Journal on Optimization*, Volume 26(2), pp. 1101–1127, 2016. *250+ citations in GS*
38. D. Telen, B. Houska, F. Logist, J. Van Impe. Multi-purpose economic optimal experiment design applied to model based optimal control. *Computers & Chemical Engineering*, Volume 94, pp. 212–220, 2016.
39. B. Houska, M.E. Villanueva, B. Chachuat. Stable set-valued integration of nonlinear dynamic systems using affine set parameterizations. *SIAM Journal on Numerical Analysis*, Volume 53(5), pp. 2307–2328, 2015.
40. B. Houska. Enforcing asymptotic orbital stability of economic model predictive control. *Automatica*, Volume 57, pp. 45–50, 2015.
41. B. Houska, D. Telen, F. Logist, M. Diehl, J. Van Impe. An economic objective for optimal experiment design of nonlinear dynamic processes. *Automatica*, Volume 51, pp. 98–103, 2015.
42. M.E. Villanueva, B. Houska, B. Chachuat. Unified framework for the propagation of continuous-time enclosures for parametric nonlinear ODEs. *Journal of Global Optimization*, Volume 62(3), pp. 575–613, 2015.

43. D. Telen, M. Vallerio, L. Cagianca, B. Houska, J. Van Impe, F. Logist. Approximate robust optimal control of nonlinear dynamic systems under parametric uncertainty and process noise. *Journal of Process Control*, Volume 33, pp. 140–154, 2015.
44. B. Houska, B. Chachuat. Branch-and-lift algorithm for deterministic global optimization in nonlinear optimal control. *Journal of Optimization Theory and Applications*, Volume 162(1), pp. 208–248, 2014.
45. D. Telen, F. Logist, R. Quirynen, B. Houska, M. Diehl, J. Van Impe. Optimal experiment design for nonlinear dynamic (bio)chemical systems using sequential semidefinite programming. *AIChE Journal*, Volume 60, pp. 1728–1739, 2014.
46. B. Houska, M. Diehl. Nonlinear robust optimization via sequential convex bilevel programming. *Mathematical Programming, Series A*, Volume 142(1), pp. 539–577, 2013.
ICCOPT Best Paper Prize in Continuous Optimization (Finalist, Top 3)
47. M. Diehl, B. Houska, O. Stein, P. Steuermann. A lifting method for generalized semi-infinite programs based on lower level Wolfe duality. *Computational Optimization and Applications*, Volume 54, pp. 189–210, 2013.
48. D. Telen, B. Houska, F. Logist, E. Van Derlinden, M. Diehl, J. Van Impe. Optimal experiment design under process noise using Riccati differential equations. *Journal of Process Control*, Volume 23, pp. 613–629, 2013.
49. B. Houska, M. Diehl. A quadratically convergent inexact SQP method for optimal control of differential algebraic equations. *Optimal Control, Applications and Methods*, Volume 34, pp. 396–414, 2013.
50. F. Logist, D. Telen, B. Houska, M. Diehl, J. Van Impe. Multi-objective optimal control of dynamic bioprocesses using ACADO Toolkit. *Bioprocess and Biosystems Engineering*, Volume 36, pp. 151–164, 2013.
51. B. Houska, F. Logist, J. Van Impe, M. Diehl. Robust optimization of nonlinear dynamic systems with application to a jacketed tubular reactor. *Journal of Process Control*, Volume 22(6), pp. 1152–1160, 2012.
52. F. Logist, M. Vallerio, B. Houska, M. Diehl, J. Van Impe. Multi-objective optimal control of chemical processes using ACADO Toolkit. *Computers & Chemical Engineering*, Volume 37, pp. 191–199, 2012.
53. B. Houska, H.J. Ferreau, M. Diehl. An auto-generated real-time iteration algorithm for nonlinear MPC in the microsecond range. *Automatica*, Volume 47(10), pp. 2279–2285, 2011. ***500+ citations in GS***
54. B. Houska, H.J. Ferreau, M. Diehl. ACADO Toolkit – An open source framework for automatic control and dynamic optimization. *Optimal Control, Applications and Methods*, Volume 32, pp. 298–312, 2011. ***1400+ citations in GS***
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1. N. Hoischen, P. Bevanda, M. Beier, S. Sosnowski, B. Houska, S. Hirche. Infinitesimal operator world models: finite-sample guarantees for offline reinforcement learning, 2026. (submitted)
2. S.K. Mulagaleti, B. Houska, M. Zanon, M.E. Villanueva. On piecewise quadratic terminal costs for MPC, 2025. (submitted; accepted for presentation at IFAC World Congress 2026)
3. F. Badalamenti, J.A. Borja-Conde, S.K. Mulagaleti, B. Houska, A. Bemporad, M.E. Villanueva. Configuration-constrained Tube MPC for periodic operation, 2025. (submitted)

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Software

1. Dr. Houska is one of the leading developers of the open-source optimal control package ACADO Toolkit. The software can be downloaded free of charge at <https://acado.github.io/>.

Miscellaneous

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